

COSIEE

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1 Creating New

New definitions and new objects can be created to solve problems, and see how they interact with the original objects.

2 Objects and Morphisms

Seek the relationship between objects and morphism.
How to turn an object into a morphism and vice versa. How to properly equip an object with morphism to make it more useful.

3 Strengthening and Weakening

We can generalize or weaken the statement to make it easier to prove.
Example: Given a group of automorphisms, we can look at the set of objects fixed by the morphism. Given a set of objects, we can look at the group of morphisms that fix them.

4 Induction

Do it one by one.

5 Extremalism

Consider EXTREME cases/objects/assumptions.

6 Equivalent perspectives

同样一个命题，我们可以从不同的视角去看待它。比如说，以下几个陈述是等价的：

$$\begin{aligned}
 & p \mid t^n - 1 \\
 & t \text{ is a root of } x^n - 1 \pmod{p} \\
 & \text{the order of } t \text{ modulo } p \text{ divides } n
 \end{aligned}$$

第一个是整出的角度，第二个是从方程的根的角度，第三个是从阶数的角度。
我们可以从不同的角度去看待一个命题。